Statistical Learning Theory
Exercise Sheet 1

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Discussion: II(c)

Within the training range $[1, 5]$, the four networks have similar errors. They are slightly less accurate than h_3 at $L = 1, 2$, but more accurate at $L = 4, 5$. At $L = 3$, the differences from h_3 are small. A larger network therefore does not consistently improve performance within this range.

For larger endpoints, the network errors increase sharply. At $L = 6$, all four networks have larger errors than h_2 and h_3. At $L = 8$, they are worse than all three comparison rules. Their MSE at $L = 8$ ranges from about 3.8 to 5.9, compared with 0.52 for h_3. The learned relationship therefore does not extend reliably to these larger endpoints.

At $L = 0.5$, performance depends strongly on the architecture. The smallest network has an MSE of about 0.004, whereas the deepest has about 0.18. Poor performance outside the training range is thus not uniform across architectures. Unlike the networks, h_1, h_2, h_3 are not fitted to the training interval. Of these three rules, h_3 is the most accurate at every tested endpoint.